



Brief Report

Exploring parent-engaged developmental monitoring of young children before and during the COVID-19 pandemic, Porter Novelli Styles 2019 and 2021

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ABSTRACT

Background: Early identification of developmental delays may have been negatively impacted by the COVID-19 pandemic. Parental engagement in developmental monitoring is a key component to successfully identifying developmental concerns.

Objective: The purpose of this project was to understand whether parental engagement in developmental monitoring changed over the course of the COVID-19 pandemic, from Spring 2019 to Fall 2021. **Methods:** Survey data were obtained from 2019 *SpringStyles* and 2021 *FallStyles* Porter Novelli Public Services *ConsumerStyles* cross-sectional surveys. Only respondents with at least one child under the age of 8 at the time of the survey were included in the analytic sample (2019 N = 403; 2021 N = 344). Participants were asked several questions about how they monitor their children's development. Changes in frequency of developmental monitoring from 2019 to 2021 were estimated using chi-squared tests.

Results: In both 2019 and 2021, 89% of parents reported engaging in any type of developmental monitoring. Within the group of parents who engaged in any monitoring, there were no differences across years in the percentage of parents reporting using the methods surveyed, except that a smaller percentage reported comparing their children to others in 2021 (25%) compared to 2019 (36%, $p < 0.002$). **Conclusions:** Despite major disruptions to families' lives, there were no significant changes to parents' overall engagement in developmental monitoring prior to and during the COVID-19 pandemic.

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One in six children between ages 3 and 17 years old have a developmental disability.¹ The American Academy of Pediatrics recommends developmental surveillance for all children, to promote early identification of and early intervention for developmental delays to improve developmental outcomes^{2–4} and lower healthcare costs.²

The effectiveness of developmental screening, a formal process using standardized, validated tools at defined ages, is well-established.^{5–18} Fewer data are available regarding the impact of developmental monitoring, which is the ongoing tracking of a child's developmental milestones by parents, teachers, and care

providers who regularly observe a child.¹⁹ Some previous work highlights the importance of developmental monitoring^{20,21} and suggests that children who receive concurrent developmental monitoring and screening are more likely to receive early intervention services compared to children receiving developmental screening alone.¹⁹ More research is needed to understand the frequency of parent-engaged developmental monitoring and referral to and receipt of services, which have been reported to be highly impacted by the COVID-19 pandemic.^{22–24}

The purpose of this project is to better understand how the COVID-19 pandemic may have impacted developmental monitoring, referral for early intervention services, and receipt of those services and supports in the United States.

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Method

We analyzed data from the 2019 *SpringStyles* and 2021 *FallStyles* Porter Novelli Public Services *ConsumerStyles* surveys. The survey is administered to a cross sectional subset of the market research firm Ipsos' KnowledgePanel® a continuously replenished panel sampled to be representative of the non-institutionalized U.S. population.²⁵ The respondents sampled in Spring 2019 were different respondents than those sampled in Fall 2021. All survey questions were the same across both time points unless noted. The data presented below were from questions that were part of a larger survey of the panel. The variables of interest did not have any missing data.

Sample

Using the weighting variables provided by Porter Novelli, samples from both surveys were weighted to match demographic proportions from the Census' American Community Survey based on the following variables: gender, age, household income, race/ethnicity, household size, education, census region, and metro status. The 2021 sample was additionally weighted by parental status of having children 12–17 years old. Developmental monitoring questions were only asked of adults with children aged < 6 years; questions about developmental concerns were only asked of adults with children aged < 8 years. In the 2019 survey, fielded May 27th to April 15th, 403 adults with a child aged < 8 years responded; in the 2021 survey, fielded from September 24th to October 7th, 344 adults with a child aged < 8 years responded.

Measures

Parents were asked "Which of the following best describes HOW you typically make sure that your child's development (how he or she speaks, moves, learns and plays) is on track for his or her age?"

Parents could select up to three choices from the following: "Compare him/her to other children the same age," "I rely on my child's doctor to tell me," "I use a checklist of developmental milestones," "I rely on my child's early care provider/teacher to tell me," "I read about child development in a book, magazine, online, etc.," "I ask my friends, family or neighbors" or "I don't do anything specific to make sure my child is on track." Parents were considered to have engaged in developmental monitoring if they chose any of the options other than that they did not do anything specific.

Three questions assessed parents' concerns about their child's development. The first question asked parents whether they or anyone else had ever had a concern about their child's development not being on track for their age. If they answered yes to this question, parents were then asked if they were ever sent by their doctor to "another professional to learn more about your child's development?" The response options were: "Yes, to a specialist (e.g., developmental pediatrician, neurologist, psychologist)," "Yes, to a speech, occupational, physical, or other therapist," "Yes, to my state's early intervention program," "Yes, to a professional or program not listed here" or "No, I was not sent to someone else for my child's development." Parents could select all responses that applied.

Finally, parents who reported ever having had a concern were asked whether their child ever received developmental support services. Parents could choose: "Yes, speech, occupational, physical or other therapy (not provided by a state early intervention program)," "Yes, services from my state's early intervention program," "Yes, services from a Head Start or other early childcare program," "Yes, services from a home visiting program," "Yes, other services not listed above," "No, my child is in the process of being evaluated to receive services," "No, my child did not receive services but

needed them," or "No, my child did not need or receive services." Parents could select all responses that applied.

Analytic approach

The unweighted frequencies and weighted percentages for all variables of interest were calculated using the R survey package.²⁶ Weights and strata were provided by Porter Novelli, more information about the calculation of these weights can be found on the Porter Novelli website.²⁷ Due to limited sample size, potential confounding by demographic or household factors could not be assessed. Pearson's Chi-squared tests were used to determine whether there was a statistically significant difference between the weighted prevalence of survey responses by year of the survey. For significant chi-squared tests with more than two groups, post-hoc comparison tests with a Bonferroni correction were conducted to determine which groups from the overall test were significantly different.

Results

In the 2019 sample of parents of children aged 8 years or younger selected from the overall *ConsumerStyles* panel, 73% of parents in the weighted sample reported having at least one child under the age of 5 years, with the remaining 27% of parents having a child aged 6–7 years old (Table 1). A significantly larger percentage of families in the 2021 sample had a child under the age of 5 years (80%) compared to the 2019 sample ($p = 0.019$). There was also a significant difference for parental age, with a smaller percentage of parents in the 18–29 years and 45–59 years categories and a larger percentage of parents in the 30–44 age range in 2021 compared to 2019 ($p < 0.001$). A greater percentage of respondents identified as Hispanic, and a smaller percentage identified as Other, non-Hispanic in 2021 compared to 2019 ($p = 0.009$). For marital status, a smaller percentage of respondents reported being married or living with a partner in 2021 while more reported never having married ($p = 0.045$). Finally, there was a smaller percentage of families who reported having a child under 1 year old in the household ($p < 0.001$) and a greater percentage who reported having a child aged 6 or 7 years in the household ($p < 0.001$) among 2021 respondents compared to 2019 respondents (see Table 2).

The weighted percentages of parents who reported engaging in developmental monitoring in the 2019 and 2021 surveys were equal (89%, unweighted N in 2019 = 249, unweighted N in 2021 = 247) in both years. The method parents most frequently reported using for development monitoring in both the 2019 and 2021 surveys was relying on a doctor to tell them (72% and 77%, respectively). Within the subsample of parents who reported engaging in developmental monitoring, there was no difference in the percentage of parents reporting using the specific methods of monitoring, except that the percentage of parents who compared their children to others of the same age was lower in the 2021 survey (25%) than in the 2019 survey (36%, $p = 0.012$).

There was no significant difference in the percentage of children whose parents indicated a concern in their child's development from the year before the pandemic to during the pandemic (Table 3). In 2019, 28% of families reported ever having a concern about their child's development; it was 26% in 2021, ($p = 0.650$). Among parents reporting a developmental concern in their child under 8 years of age, a similar percentage of families reported being sent by their provider to another professional to learn more about their child's development (62% in 2019 vs 61% in 2021; $p = 0.720$). Among parents who reported a concern, similar percentages reported that their child is receiving services to support their development (54% in 2019 vs 50% in 2021; $p = 0.566$).

Table 1
Demographic characteristics of parents of children < 8 years, 2019 *SpringStyles* and 2021 *FallStyles*.

Characteristic	Spring 2019 (n = 403)		Fall 2021 (n = 344)		p-value ^b	Post-hoc p-values
	% ^a	N	%	N		
Parental age						
18–29	36%	100	20%	40	0.001	0.001
30–44	57%	259	66%	247		0.017
45–59	6%	38	13%	47		0.005
60+	1%	6	2%	10		1.000
Child age						
0–5 years	73%	283	80%	272	0.019	
6–7 years	27%	120	19%	72		
Race/Ethnicity						
White, non-Hispanic	59%	263	60%	232	0.009	1.000
Black, non-Hispanic	14%	41	10%	20		0.882
Hispanic	10%	28	18%	45		0.001
Other, non-Hispanic	17%	56	11%	36		0.060
2+ races, non-Hispanic	1%	15	2%	11		1.000
Marital Status						
Married/living with a partner ^c	85%	348	76%	284	0.045	0.001
Never married	8%	24	15%	33		0.001
Widowed/divorced/separated	7%	31	9%	27		1.000
Education						
Less than high school	10%	21	10%	16	0.225	
Completed high school	27%	86	26%	72		
Some college	19%	87	26%	83		
Bachelor's degree+	45%	209	38%	173		
Employment status						
Unemployed	23%	306	28%	76	0.257	
Employed	77%	97	72%	268		
Household income						
< \$25,000	16%	52	12%	23	0.477	
\$25,000 – <\$50,000	19%	69	17%	44		
\$50,000 – <\$100,000	31%	135	33%	114		
\$100,000 – <\$150,000	18%	72	17%	67		
\$150,000+	17%	75	21%	96		
Households with members in this age group ^d						
0–1 years	46%	182	31%	96	0.001	
2–5 years	66%	268	68%	234	0.596	
6–7 years	28%	120	41%	148	0.001	

^a Percents are weighted, Ns are unweighted.

^b Pearson's Chi-squared test. Chi-square p-value significant < 0.05, post-hoc tests use Bonferroni correction.

^c Living with a partner was not a response option in the 2019 survey and was combined with the married category in 2021.

^d Ns in this category will sum to more than the total sample because families may have multiple children, percentages will sum to more than 100% for the same reason. Chi-squared tests were conducted within each age group.

Table 2
Among parents with children < 6 years who reported developmental monitoring, percentage of parents using various methods for conducting developmental monitoring, 2019 *SpringStyles* and 2021 *FallStyles*.

How do you typically make sure that your child's developmental (speech, movement, learning, and playing) is on track for his/her age?*	Spring 2019 (n = 249)		Fall 2021 (n = 247)		p-value ^b
	% ^a	N	%	N	
Compare him/her to other children the same age	36%	95	25%	68	0.012
I rely on my child's doctor to tell me	72%	179	77%	189	0.414
I use a checklist of developmental milestones	29%	74	34%	87	0.258
I rely on my child's early care provider/teacher to tell me	33%	85	34%	89	0.875
I read about child development in a book, magazine, online, etc.	35%	89	33%	81	0.557
I ask my friends, family, or neighbors	22%	57	19%	41	0.473

^a Percents are weighted, Ns are unweighted.

^b Chi-squared tests on weighted Ns, p-value significant < 0.05.

Discussion

Parental engagement in developmental monitoring is an integral component of ensuring developmental concerns for young children are detected and addressed early.⁶ The COVID-19 pandemic led to dramatic alterations in the lives of families and understanding how these changes impacted parents' developmental monitoring can help developmental monitoring programs and healthcare providers support parents as advocates for their children.

These survey data suggest that engagement in parental monitoring was not different in the year before the pandemic (Spring of 2019) to approximately a year and a half into the pandemic (Fall of 2021) and aligns with previously published estimated rates of developmental monitoring.²¹ While there was no difference in the percentage of parents engaging in monitoring overall, within the group of parents who engaged in any developmental monitoring a significantly smaller percentage reported that they compared their children to other children of the same age during the pandemic compared to the year before the pandemic. This difference may in

Table 3

Percentage of parents with children < 8 years reporting developmental concerns, referral for services, and receipt of services for early identification of developmental disabilities, 2019 *SpringStyles* and 2021 *FallStyles*.

Survey Question	Response	Spring 2019		Fall 2021		p-value ^b
		% ^a	N	%	N	
Thinking about your child(ren) under age 8, have you or anyone else ever had a concern about their development not being on track for their age?	Yes	27%	111	26%	83	0.650
	No	73%	291	74%	261	
For the child there were concerns about, did your child's doctor or other provider ever send you to another professional to learn more about your child's development?	Yes	62%	74	61%	47	0.720
	No	38%	37	39%	36	
Did the youngest child there were concerns about receive services to support his/her development?	Yes	54%	65	50%	37	0.566
	No	46%	46	50%	46	

^a Percents are weighted, Ns are unweighted.

^b Chi-squared tests on weighted Ns, p-value significant <0.05.

part be due to school closures and public health recommendations to physically distance from people outside the household. Given the potential benefits of using developmental checklists, including increasing caregiver's awareness of developmental milestones and discussions with pediatricians, parents less frequently engaging in less rigorous forms of developmental monitoring such as comparing their children to their peers may help improve identification of developmental delays.²⁸

The most frequently reported method of developmental monitoring reported by parents at both time periods was relying on a doctor to tell them (72% and 77%). However, as those who likely spend the most time with their children, parents may be better suited to identify a developmental concern earlier than medical professionals who may only see children once or twice a year.³ There were also no significant differences in the percentage of parents who reported using resources for developmental monitoring (i.e., books, magazines, online webpages, etc.), in the year before and during the pandemic. This finding suggests that publicly available resources, such as those developed and made widely available by CDC and other programs, were being utilized during the COVID-19 pandemic. There were also no statistically significant differences in the percentage of parents using a checklist of developmental milestones in the 2019 and 2021 surveys. However, in both surveys only about 30% of parents reported using a checklist. These results suggest possible areas of intervention, as many parents may not be aware of the advantages of engaging in developmental monitoring themselves, as opposed to relying on medical professionals, or the tools available to aid them in monitoring, such as milestone checklists.

The percentage of parents reporting a concern for their children's development did not significantly differ from 2019 to 2021. While this finding is reassuring, it should be interpreted cautiously given widespread concern that children's mental health, social-emotional learning and communication skills, have all been negatively impacted by the unpredictability of pandemic-related disruptions to families' lives.^{29–34} It is possible that negative impacts on children's development may only become clear to both families and researchers further into the future or that the types of developmental monitoring parents are engaging in are missing early concerns. Future work will need to examine how children's development has been impacted, as children who might have been identified as needing intervention in 2020 when lockdowns were prevalent, continue to develop. Multiple theories of child development state that reduced exposure to a variety of environments and stimuli may have a limiting effect on children's development and that increased parental stress may have negative impacts on children.³⁵ To address this potential future issue, parents can be made aware of what signs and symptoms to look out for as their children grow, to identify and address potential issues as early as possible. Among families who had concerns about their child's development, the percentage who were referred to other

professionals or received services also did not differ from the year before to during the pandemic, a trend that differs from some European countries.²³ Concerningly, approximately half of the parents who expressed a developmental concern for their child did not report receiving services to support their child's development at either timepoint.^{35,36}

Data on how parent engagement in developmental monitoring were impacted by the COVID-19 pandemic in U.S. populations is lacking. It is important to understand the extent to which parents may need more support in engaging in developmental monitoring, especially given the reduced number of contact points families may have had with healthcare and educational professionals during the pandemic. The sample used in this study provides a snapshot of parent-initiated developmental monitoring, referral for early intervention services, and receipt of early intervention services in a sample of U.S. families with children.

The current study had several limitations. While we assessed differences over time, the survey did not sample respondents longitudinally. Comparing the two groups by their demographic characteristics revealed that there were some significant differences between the 2019 and 2021 samples. Although few differences in developmental monitoring were noted between the 2019 and 2021 surveys, sample size was insufficient to assess whether differences in monitoring variables may have existed in subgroups defined by demographic and other characteristics, such as differences by children's age which was likely an important factor in parental monitoring during the COVID-19 pandemic. In addition, the sample was recruited by a market research firm as part of a panel survey and participants were compensated for their time. The sample was not recruited to identify parents of young children and respondents agreed to participate in the panel on an opt-in basis until the survey quota was reached rather than randomly sampled. These issues limit the generalizability of the findings to the population of parents in the U.S. Finally, the survey questions asking about developmental concerns and referral to or receipt of services did not refer to a specific point in time, parents were asked if these things had ever happened. It is therefore possible that parents in the 2021 survey were responding about experiences that had happened prior to the pandemic.

While the COVID-19 pandemic led to disruption for children with disabilities and their families, there may now be an opportunity to reshape systems so that they will better support the most vulnerable members of society.³⁷ Understanding how families have been impacted by the COVID-19 pandemic is a step to identifying areas in which parents need greater support to identify and address developmental concerns for their children. The high level of reliance on medical professionals to identify developmental delays and the underutilization of developmental monitoring tools designed for parents suggest an opportunity for medical professionals to promote the use of these tools to engage parents in monitoring and support conversations about possible developmental concerns.

Disclaimer

The contents of this article do not necessarily represent the official position of the Centers for Disease Control and Prevention.

Conflict of interest

The authors have no conflicting interests to report.

Submission declaration

The work described here is not currently under consideration for publication nor has it been published elsewhere.

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